

## **Conditional Volatility of AEX Returns:**

### *A Comparison of GARCH and GJR-GARCH Models*

Authors:

Maxim Milde (73267075@fsv.cuni.cz)

Zahid Pashayev (52520099@fsv.cuni.cz)

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### **Abstract**

This paper examines the time-varying nature of conditional volatility of daily log-return observations on the Amsterdam Exchange Index (AEX), from January 4, 2010, through March 6, 2023. The symmetric GARCH (1,1) model is employed to evaluate the respective contributions of mean-reversion and volatility persistence in the AEX return process under two different distributional assumptions: normal and student-t. Following rejection of the normality assumption via various information-criteria-based diagnostics and specific return-series based diagnostic tests, an asymmetric GJR-GARCH (1,1) model was employed under student-t innovations. Our empirical results show that there are large clusters of volatility, that the ARCH effect is highly significant, and that the return series exhibits excess kurtosis. Sign-bias diagnostic results demonstrate that volatility increases more after negative return shocks than positive ones of similar magnitude; this motivated our use of an asymmetric GARCH model. Results also indicate that, based on information criterion values and a likelihood ratio test, the student-t GJR-GARCH (1,1) is the best-fitting model. These findings support previous leverage-effect research and emphasize the necessity of accounting for fat tails and asymmetry when modeling the volatility associated with equity index returns.

**Keywords:** Conditional volatility, GARCH, GJR-GARCH, TARCH, leverage effect, AEX, Student- $t$  distribution, volatility clustering.

**JEL Classification:** C22, C58, G10.

## 1. Introduction

Estimating the volatility of returns from financial assets and predicting them is a fundamental objective of empirical finance. The accuracy of volatility estimates is crucial for managing risks, evaluating derivatives, and allocating funds to portfolios. Traditional economic modeling approaches, for example, ordinary least squares (OLS), and autoregressive integrated moving average (ARIMA) frameworks assume that the variance of a time series does not vary over time. In contrast, this assumption is often broken in the case of financial data due to volatility clusters, i.e., when several large price movements occur at approximately the same time - an empirical regularity described as volatility clustering (Cont, 2001).

In response to this issue, Engle (1982) developed the Autoregressive Conditional Heteroscedasticity (ARCH) model allowing the conditional variance to be influenced by prior squared residuals. Bollerslev (1986) further extended ARCH with the inclusion of lagged conditional variances to produce the Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model. GARCH has become one of the most widely used techniques to capture the persistence of volatility patterns observed within financial markets.

Asymmetric GARCH Models are also subject to limitations. Symmetric GARCH models are limited in their ability to assign a greater or lesser weight to both positive and negative return shocks of the same size. However, empirical evidence has shown that bad news (or negative returns) can increase future volatility at a faster rate than good news (or positive returns) of the same size. This phenomenon is called the Leverage Effect (Black, 1976). To identify these types of asymmetries, Engle and Ng (1993) developed methods for testing for sign-bias. Similarly, Glosten et al. (1993) introduced the GJR-GARCH Model (also known as TAR-CH Model) as a means of identifying whether negative shock impacts were larger than those of positive shocks; this was done by incorporating an indicator variable into the Variance Equation.

In this paper, the GARCH and GJR-GARCH models are applied to the daily closing price of the Amsterdam Stock Market Index (AEX) from January 4th, 2010, to March 6th, 2023. The AEX index is a widely used Dutch Equity-Benchmark, and it consists of the 25 most actively traded equities listed on the Euronext Amsterdam exchange. Therefore, our sample includes all major periods of market turbulence during the past decade including the European Sovereign Debt Crisis (2010-2012), the post-crisis recovery period (2013-2019) and the COVID-19 induced stock-market crash/rebound (2020-2021).

Therefore, the primary research question to which this evaluation is directed is as follows: Does the sign of return shocks have implications for conditional volatility estimates in the AEX index; and if it does, will the GJR-GARCH (TAR-CH) Model better capture these dynamics than the traditional GARCH (1,1)?

To evaluate these questions, we test four null hypotheses. The first hypothesis addresses whether the conditional volatility of AEX log-returns is constant over time (no ARCH effects). The second hypothesis states that both positive and negative shocks of an identical size will symmetrically affect the conditional volatility. The third hypothesis states that the GJR-GARCH (1,1) specification does not produce a better fit to the conditional volatility process. The fourth hypothesis states that the normal distribution provides a reasonable description of the innovations process. We implement an iterative strategy from bottom to top to address these issues. We begin by evaluating the mean equation. Then, we test for ARCH effects. Next, we estimate both symmetric and asymmetric GARCH specifications. Finally, we compare the performance of each model specification using information criteria, LR tests, and other diagnostic tests. The rest of this paper is organized as follows:

## 2. Literature Review

The ARCH model of Engle (1982) was the first econometric specification that formalized allowing for a time series' conditional variance to vary as a function over time based on past squared residuals. This represents significant improvement over constant-variance models; however, pure ARCH models require many lags to capture volatility persistence typically observed in financial markets, which results in high-dimensional parameter vectors. Bollerslev (1986) addressed the shortcomings of pure ARCH by proposing the generalized ARCH (GARCH) model, which includes lagged conditional variances in the variance equation. The GARCH (1,1) with one ARCH lag and one GARCH lag has become the standard specification used in applied finance due to its parsimony and ability to model slow decay rates for autocorrelations of volatility.

There is a stylized fact present within equity markets, that the volatility response to return shocks is asymmetric. Negative returns tend to increase future volatility more than positive returns of equal absolute magnitude. Black (1976) offered an early explanation of this phenomenon based upon leverage, when a firm's equity value falls its debt-to-equity ratio increases making the stock riskier. However, research since has demonstrated that the empirical magnitude of the asymmetry exceeds what can be explained alone through leverage feedback channel mechanisms, suggesting risk-premium feedback and behavioral mechanism also play a role (Christie, 1982).

Several GARCH models that allow for asymmetry have been developed; among these, Nelson's (1991) EGARCH model is perhaps most like the GJR-GARCH model considered in this paper. A primary motivation for focusing on the GJR-GARCH model lies in its ability to test directly whether negative price movements increase conditional variance more than positive price movements.

As far as the distributional assumptions regarding the innovations process of GARCH models, Bollerslev (1987) found that many times when estimated GARCH models assume the error terms follow a normal distribution, standardized residuals exhibit excess kurtosis. This finding implies that the assumed normal distribution may not be fat tailed enough. As such, Bollerslev (1987) recommended estimating GARCH models based upon a student-t distribution. The student-t distribution incorporates a shape parameter that determines the degree of "fatness" at both tails of the distribution.

This study builds upon existing literature by utilizing a generalized form of GARCH (i.e., GARCH (1,1)) with normal and student-t innovations and GJR-GARCH (1,1) using student-t innovations for modeling daily returns in AEX. The primary purpose of our study is educational as it illustrates a full model development process that includes stationarity testing, identifying an appropriate ARMA specification for the conditional mean equation, detecting the presence of ARCH effects, and comparing models via different information criteria.

## 3. Data Description

The dataset for this research includes each day's closing price of the AEX Index. Prices are obtained via Yahoo Finance in the period spanning from January 4th, 2010, to March 6th, 2023. There is a total of 3375 trading days in our data set. One day (January 2nd, 2012) has a missing closing price and is therefore removed. This leaves us with 3374 remaining prices. Continuously compounded returns are computed using  $r_t = \ln(P_t) - \ln(P_{t-1})$ , where  $P_t$  represents the closing price on day  $t$  and  $P_{t-1}$  represents the closing price on the previous day. Therefore, there are a total of 3373 daily log-return observations.

**Table 1: Descriptive Statistics of AEX Daily Log>Returns**

| Statistic             | Value      |
|-----------------------|------------|
| Observations          | 3,373      |
| Mean                  | 0.000236   |
| Standard deviation    | 0.011325   |
| Minimum               | -0.11376   |
| Maximum               | 0.08591    |
| Skewness              | -0.500     |
| Excess kurtosis       | 6.759      |
| Jarque-Bera statistic | 6,561.0*** |

Note: \*\*\* denotes rejection of the null hypothesis of normality at the 1% significance level.

Table 1 presents summary statistics of the main characteristics of the log-return time series. The average log-return over each trading day is very small (close to 0.024%) as expected from the typical near-martingale behavior seen for one-day stock returns. The standard deviation of the log-returns at the daily frequency (1.13%) can be transformed into the corresponding annualized volatility by multiplying it by  $\sqrt{252} = 15.87$ . This results in an approximate value of 18.02%, which indicates high uncertainty in daily stock prices. The return series has a left-skewed distribution (-0.5). Therefore, large losses occur much more often than their positive counterparts of the same size. The excess kurtosis of 6.76 is greater than 0 and thus confirms a fat-tailed return-distribution characteristic of stock price movements.



Figure 1: AEX Closing Price (January 2010 - March 2023)

Figure 1 depicts the AEX closing price for the entire sample period. The AEX had a net positive trend from about 340 at the beginning of 2010, until it reached approximately 760 at the end of February 2023. There were major corrections and drawdowns throughout this time due to the European Sovereign Debt Crisis (2011-12), a China driven correction (late 2015 to early 2016), and most notably the sharp selloff caused by the COVID-19 pandemic in March of 2020.

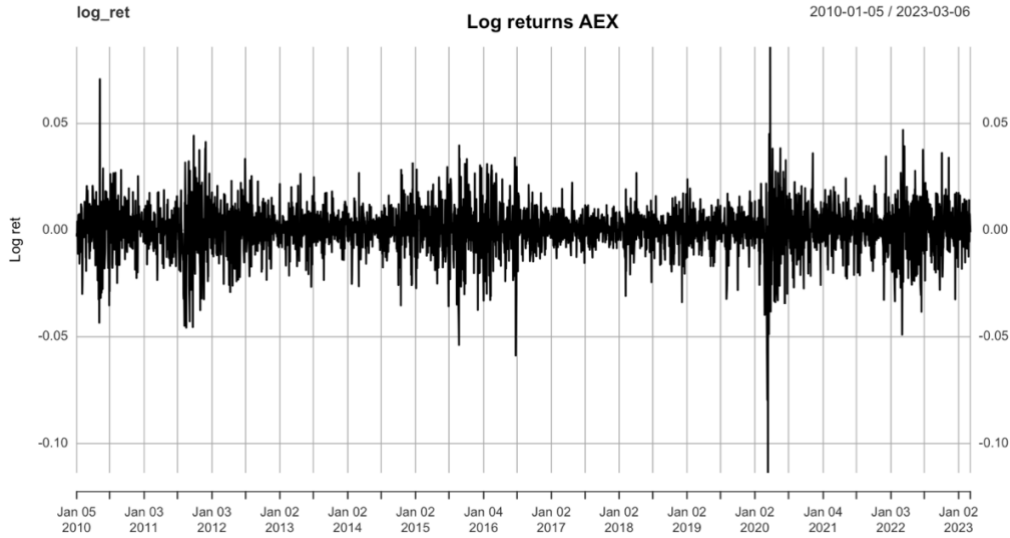


Figure 2: AEX Daily Log-Returns (January 2010 - March 2023)

Figure 2 displays the daily log-return series. The plot reveals pronounced volatility clustering: periods of elevated return variation (notably 2011 to 2012 and 2020) are followed by extended calm phases - a visual signature of conditional heteroskedasticity.

## 4. Empirical Methodology

### 4.1 Stationarity Testing

Before we attempt to estimate our models, we first assess whether both time series are stationary in terms of their behavior over time. To accomplish this, we use two different types of unit-root/stationarity tests. The Augmented-Dickey Fuller (ADF) tests against the presence of a unit root (i.e., non-stationarity), whereas the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests for stationarity. Using both tests together gives a more reliable picture.

### 4.2 Mean Equation Specification

We use an ARMA  $(p, q)$  to represent the conditional mean of the returns data. The first step in model selection is to apply the automated method of Hyndman and Khandakar (2008). This method selects the best model by minimizing the BIC across all possible order combinations through a grid search. We also manually estimate ARMA (1,1), ARMA (2,2), and ARMA (5,5) to check for overparameterization and coefficient redundancy. To verify whether our selected model has residual autocorrelation, we examine it through both autocorrelation function/partial autocorrelation function patterns and the Ljung-Box Portmanteau test. If after these tests no ARMA components are able to provide any additional value as predictors of returns, we will select a model representing constant mean ARMA (0,0), essentially de-meaning the returns data.

### 4.3 Testing for ARCH Effects

We then need to determine if our residuals display evidence of time-dependent variability; there are several methods that can be used to accomplish this. The first method is to examine the ACF/PACF of the squared residuals. If the residuals show clustering of volatility, it should be evident in the ACF/PACF as well. Second, we perform an ARCH-LM test by running a regression of the squared residuals against lagged values of the squared residuals and performing an F-test to see if the coefficient estimates are statistically different from zero. This test is developed by Engle (1982). The

third option is to use a Ljung-Box test for the squared residuals to find out if any residual autocorrelation remains. If any or all these tests reject the null hypothesis that the variances remain constant over time, we would reject  $H_{0a}$  and proceed with estimating the parameters of a GARCH model.

#### 4.4 GARCH (1,1) Model

Our starting model for analysis is the GARCH (1,1), whose variance equation follows the form:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$$

where the term  $\omega$  represents the long run variance. The parameter  $\alpha$  tells us how much of yesterday's shock to the mean influences today's variance. This is called the ARCH effect. The coefficient  $\beta$  measures persistence in volatility - How much of yesterday's variance "carries forward" into today. If the parameters  $\alpha + \beta$  are near 1 then it can take a long time for volatility shocks to decay, consistent with our observations on most financial data. To verify the order choice, we conduct a BIC grid search over GARCH ( $p, q$ ) for  $p, q$  up to 5.

#### 4.5 GJR-GARCH (1,1) Model

The Symmetric GARCH Model treats all shocks identically, whether they are positive or negative. To account for asymmetry of shocks, we estimate a GJR-GARCH (1,1) of Glosten, Jagannathan, and Runkle (1993). It is like the classic symmetric GARCH model but incorporates an additional term to the variance equation. Specifically:

$$h_t = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta h_{t-1}$$

This modification includes the use of the dummy variable  $I_{t-1}$ . The dummy variable is equal to zero if yesterday's shock was positive, and 1 otherwise. Therefore, the conditional variances will be affected differently by shocks based upon their sign. Negative shocks will have an effective ARCH coefficient  $(\alpha + \gamma)$  compared to positive shocks  $(\alpha)$ . If  $\gamma$  is statistically significant and greater than zero, then the impact of bad news (negative shocks) increases volatility disproportionately in comparison to positive shocks. If  $\gamma = 0$ , then the model reduced to the standard GARCH (1,1) which is our null hypothesis  $H_{0b}$ . Volapersistence in this model is given by  $\alpha + \beta + \gamma/2$ ; and since these models are nested, we can perform a formal comparison via a likelihood ratio test using one degree of freedom.

#### 4.6 Distributional Assumptions & Estimation

We estimate two versions of the symmetrical GARCH (1,1): one version assumes the standardized residuals are normally distributed, and the other version assumes the standardized residuals are student-t distributed. The student-t has an additional parameter  $\nu$  which can control the heaviness of the tail of the distribution. As  $\nu$  gets smaller, the tails get heavier, and as  $\nu$  increases toward infinity, the student-t distribution approaches normality. After determining normality cannot be assumed (using both information criteria and residual diagnostic tests), we then run the GJR-GARCH (1,1) model using student-t distributions for the residuals. Given the evidence against normally distributed innovations, the asymmetric specification is estimated using student-t innovations only. We use Maximum Likelihood estimation for all models using the *rugarch* R package (Ghalanos, 2024) and report results from both standard Maximum Likelihood Estimation (MLE) errors and the robust version of Bollerslev and Wooldridge (1992).

## 4.7 Model Comparison and Diagnostics

To determine which of the two models better describes our data, we first evaluate each model based on four different information criteria - Akaike Information Criteria (AIC), Bayesian Information Criterion (BIC), Shibata's criterion, and the Hannan-Quinn (HQ) information criterion. For nested models, we have one additional measure to compare models by using the Likelihood Ratio Test. To assess this aspect of model evaluation, we can use three residual diagnostics. The Ljung-Box test evaluates whether there is significant correlation present within both standardized residuals and the squared values of the residuals. The ARCH-LM test evaluates whether there is any remaining heteroscedasticity left over in the residuals. The Sign Bias test, originally proposed by Engle and Ng (1993), examines whether the residuals continue to exhibit differential responses to positive versus negative shocks. As part of our overall model evaluation process, we also compute Jarque-Bera statistics, perform Nyblom stability tests, review QQ-plots and histogram plots of the residuals to check how closely the actual residuals resemble those from a normal distribution. Lastly, we compute a news impact curve (which relates the contemporaneous shock to next period variance) so that we may visualize any remaining asymmetries in the model.

## 5. Results & Discussion

### 5.1 Stationarity

For the AEX price level, the ADF and KPSS tests provide conflicting evidence. Since equity index levels are typically highly persistent and the KPSS test rejects stationarity, the analysis proceeds with log-returns. As indicated by both the high persistence of equity index levels and by the KPSS test results, we perform our analysis using returns in logarithm form. For the log-return series, the ADF rejects its null of a unit root, and the KPSS fails to reject its null of stationarity, jointly providing strong evidence that the return series is stationary.

*Table 2: Unit Root and Stationarity Tests*

| Series          | ADF Result                   | KPSS Result                  | Conclusion                                  |
|-----------------|------------------------------|------------------------------|---------------------------------------------|
| AEX Price Level | Rejects H0 (borderline)      | Rejects H0 ( $p < 0.01$ )*** | Conflicting; analysis proceeds with returns |
| AEX Log-Returns | Rejects H0 ( $p < 0.01$ )*** | Fails to reject H0           | Stationary                                  |

*Note: ADF null: series has a unit root. KPSS null: series is stationary. \*\*\* significant at the 1% level.*

### 5.2 Mean Equation Selection

To select an appropriate conditional mean equation, we employ both the BIC based `auto.arima()` function as well as our own analysis of various manually specified ARMA (1,1), ARMA (2,2), and ARMA (5,5) specifications. In all cases, the ARMA (0,0) model with a constant mean provides the smallest BIC. Our manual examination reveals that higher order ARMA models are producing pairs of nearly cancelling AR and MA parameters which does not enhance the quality of the residuals relative to the simpler models.

### 5.3 ARCH Effects

There are long periods with little activity punctuated by large jumps, particularly during the sovereign debt crisis from 2011 - 2012 and again with the COVID-19 pandemic in 2020. This pattern is further illustrated through the plots of both the auto-correlogram (ACF) and partial-auto correlogram (PACF) for the squared residuals where there is evidence of autocorrelation in the data

at numerous lags which illustrates that the volatility or variance of stock price returns is changing over time. The ARCH-LM test and Ljung-Box test on squared residuals both reject their null hypotheses at  $p < 0.001$ , confirming strong ARCH effects. This justifies estimating a GARCH-type volatility model. We reject  $H_{0a}$  and conclude that conditional volatility is not constant over time.

**Table 3: GARCH (1,1) - Normal Distribution**

| Parameter | Estimate | Std. Error | Robust S.E. | t-stat (robust) |
|-----------|----------|------------|-------------|-----------------|
| mu        | 0.000536 | 0.000122   | 0.000139    | 3.858***        |
| omega     | 0.000002 | 0.000001   | 0.000001    | 2.459**         |
| alpha     | 0.0878   | 0.0115     | 0.0173      | 5.075***        |
| beta      | 0.8991   | 0.0129     | 0.0176      | 51.08***        |

Note: \*\*\* significant at 1%, \*\* at 5%. Robust standard errors from Bollerslev-Wooldridge

The results from our GARCH (1,1) show that all four coefficients are significant according to both the Maximum Likelihood Estimates (MLE), as well as the robust standard errors reported in table 3. the ARCH coefficient  $\alpha = 0.088$  indicates that approximately 9% of the previous day's squared shock will affect the current day's variance; this is a high degree of short run response.

The GARCH coefficient  $\beta = 0.899$  shows an even greater degree of persistence. Taken together,  $\alpha + \beta = 0.987$  represents an almost perfect unit root process indicating that once volatility has spiked, it will take a very long time to recede. In addition to estimating the GARCH parameters, we also estimate the annualized long-run volatility, which is close to 17.8%. This estimate is consistent with the sample standard deviation of returns. A BIC grid search over GARCH (p, q) specifications through order five concludes that the best specification for a symmetric model is GARCH (1,1)

We conducted the Ljung-Box test and ARCH-LM test at various lag levels, where neither rejected the null hypothesis. Such results indicate that the model is capturing most of the variance dynamics of the data. However, there are two remaining issues. First, the sign bias test of Engle and Ng (1993) produces evidence that indicates a statistically significant ( $p < 0.01$ ) negative sign-bias coefficient - negative shocks appear to impact next period's variance more strongly than positive shocks; and since the GARCH variance equation is assumed to be symmetric, it cannot account for this phenomenon. Second, we also observe rejection of the Goodness-of-Fit test for normality for standardized residuals; in particular, they exhibit kurtosis of around 4.2, whereas the expected value under normality is 3.0. Although we fit a GARCH model, our standardized residuals continue to exhibit tails that are much heavier than those assumed under normality.

#### 5.4 GARCH (1,1) with Student-t Errors

**Table 4: GARCH (1,1) — Student-t Distribution**

| Parameter | Estimate | Std. Error | Robust S.E. | t-stat (robust) |
|-----------|----------|------------|-------------|-----------------|
| mu        | 0.000524 | 0.000112   | 0.000130    | 4.026***        |
| omega     | 0.000002 | 0.000001   | 0.000001    | 2.07**          |
| alpha     | 0.0836   | 0.0128     | 0.0198      | 4.22***         |
| beta      | 0.9038   | 0.0139     | 0.0196      | 46.11***        |
| shape (v) | 7.27     | 0.68       | 0.94        | 7.73***         |

Note: \*\*\* significant at 1%, \*\* at 5%. Shape is the degrees-of-freedom parameter of the student-t distribution.

Replacing the normal distribution with a student-t distribution results in an improvement in the fit of the data, although it does very little to alter the variance equation parameters. The shape parameter for the student-t distribution is approximately 7.3, low enough to imply heavier tails than the normal distribution, while still allowing finite variance. The values of both  $\alpha$  and  $\beta$ , however, are very close to their respective values when using a normal distribution; this is consistent with quasi-maximum-likelihood estimation (QMLE) theory: while one can get inefficient estimates if the assumption of the distribution is wrong, they will still be consistent.

The student-t distribution performs better than the normal distribution when performing model selection, as it outperforms in all four information criteria used: AIC, BIC, Shibata's criterion, and Hannan-Quinn criterion. We reject  $H_0$  because the normal distribution could not possibly account for the extreme returns in our sample.

**Table 5: Information Criteria — Model Comparison**

| Criterion    | GARCH (1,1)<br>Normal | GARCH (1,1)<br>Student-t | GJR-GARCH (1,1)<br>Student-t |
|--------------|-----------------------|--------------------------|------------------------------|
| AIC          | -6.3641               | -6.3862                  | -6.3903                      |
| BIC          | -6.3568               | -6.3771                  | -6.3793                      |
| Shibata      | -6.3641               | -6.3862                  | -6.3903                      |
| Hannan-Quinn | -6.3613               | -6.3828                  | -6.3862                      |

Note: Lower values indicate better fit. The GJR-GARCH (1,1) with Student-t errors achieves the best (lowest) value on all four criteria.

## 5.5 GJR-GARCH (1,1) with Student-t Errors

**Table 6: GJR-GARCH (1,1) — Student-t Distribution**

| Parameter       | Estimate | Std. Error | Robust S.E. | t-stat (robust) |
|-----------------|----------|------------|-------------|-----------------|
| mu              | 0.000535 | 0.000110   | 0.000126    | 4.25***         |
| omega           | 0.000002 | 0.000001   | 0.000001    | 2.14**          |
| alpha           | 0.0225   | 0.0145     | 0.0159      | 1.41            |
| beta            | 0.9063   | 0.0137     | 0.0177      | 51.21***        |
| gamma           | 0.1097   | 0.0234     | 0.0370      | 2.97***         |
| shape ( $\nu$ ) | 7.63     | 0.74       | 1.04        | 7.33***         |

Note: \*\*\* significant at 1%, \*\* at 5%.  $\gamma$  is the asymmetry coefficient on the indicator  $I(\epsilon < 0)$ . Under robust SE,  $\alpha$  is not individually significant.

We are confident that the model captures the asymmetry effect because the asymmetry parameter  $\gamma = 0.110$  is positive and statistically significant at the 1 percent level under both standard ML standard errors and robust standard errors (robust  $t = 2.97$ ). Therefore, the model clearly shows that negative shocks affect volatility significantly more than positive shocks. Our results show that this difference is large; that is the estimated ARCH coefficients for yesterday's shock being negative are  $(\alpha + \gamma) = 0.0225 + 0.1097 = 0.1322$ . The ARCH coefficient for yesterday's shock being positive is just  $\alpha = 0.0225$ . The point estimates indicate that short run volatilities in response to negative shocks are about 6 times greater than those of positive shocks.

What also stands out from these estimates is how insignificant  $\alpha$  alone is using robust standard errors ( $t = 1.41, p = 0.16$ ). What this means is that even after allowing for an additional pathway through which negative shocks may cause increased volatility for future periods, positive shocks do not have a statistically significant independent effect on next-period variance once asymmetry is included. Thus, overall, we see two sources of variation driving this volatility process: (i) the persistence of past conditional variance itself ( $\beta = 0.906$ ); and (ii) the disproportionately large impact of negative shocks ( $\gamma = 0.110$ ). The overall persistence of variance in the system ( $\alpha + \beta + \gamma / 2$ ) = 0.984, remains very close to unity and comparable to a symmetric GARCH meaning volatility shocks take equally long to disappear in both models; however, with a symmetric GARCH, new variance arises from shocks of either sign and the short-run shock response is mainly driven by negative shocks.

A final piece of evidence supporting the rejection of the null hypothesis is provided by a likelihood ratio test. The test compares the nested GARCH (1,1)-t to the GJR-GARCH (1,1)-t and rejects the symmetric restriction at the 1 percent level ( $p < 0.001, df = 1$ ). As such, this is stronger than what could be gleaned from information criteria tests and constitutes direct statistical evidence for rejecting  $H0c$ .

The News Impact Curve (see Figure 4) illustrates this asymmetry. The symmetric GARCH generates a symmetric U-shaped news impact curve centered around zero. Hence, a 2 percent decline and a 2 percent increase result in identical responses in variance. On the other hand, the left side of the GJR-GARCH has a much steeper slope than the right. Most intuitively, this represents the leverage effect, i.e., negative shocks generate larger changes in the variance process.

In terms of diagnostics, the Ljung-Box test fails to reject the null hypothesis of no first-order autocorrelation in both the standardized residuals and their squares, which indicates no remaining autocorrelation in either the mean or the variance equation. The ARCH-LM test similarly indicates that there exists no unaccounted for heteroskedasticity.

Combining all results together, the GJR-GARCH (1,1) with Student-t innovations has the smallest values for any information criterion calculated (see Table 5); the model passes both the main residual autocorrelation and remaining ARCH tests (also confirmed via a significant Likelihood Ratio Test). Additionally, our results show some indication of parameter instability in  $\omega$  as indicated by the Nyblom test which may be due to changes in variance during crisis periods. Therefore, our results support rejecting  $H0c$ : specifically, we have demonstrated that the asymmetric model provides a significantly better fit than the symmetric GARCH (1,1).

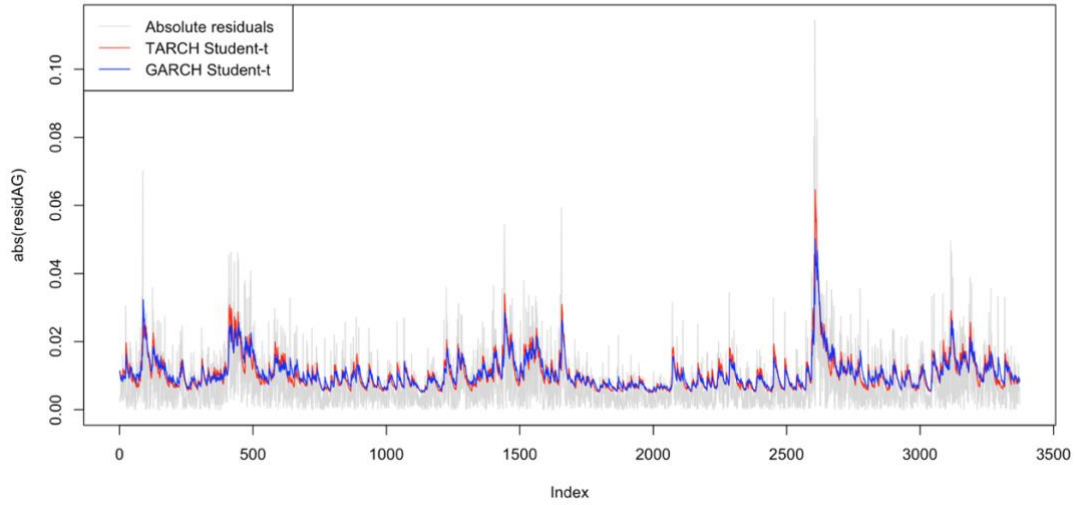


Figure 3: Conditional Standard Deviation: GARCH (1,1)-t vs GJR-GARCH (1,1)-t

The estimated conditional standard deviation of GARCH (1,1)-t model versus the estimated conditional standard deviation of GJR-GARCH (1,1)-t model are shown in Figure 3. The two series follow a very similar pattern, however, the GJR-GARCH model produces slightly larger volatility estimates for periods when the AEX index experienced large declines such as the period of March 2020. This reflects how the GJR-GARCH model responds differently to shocks depending on whether they are positive or negative.

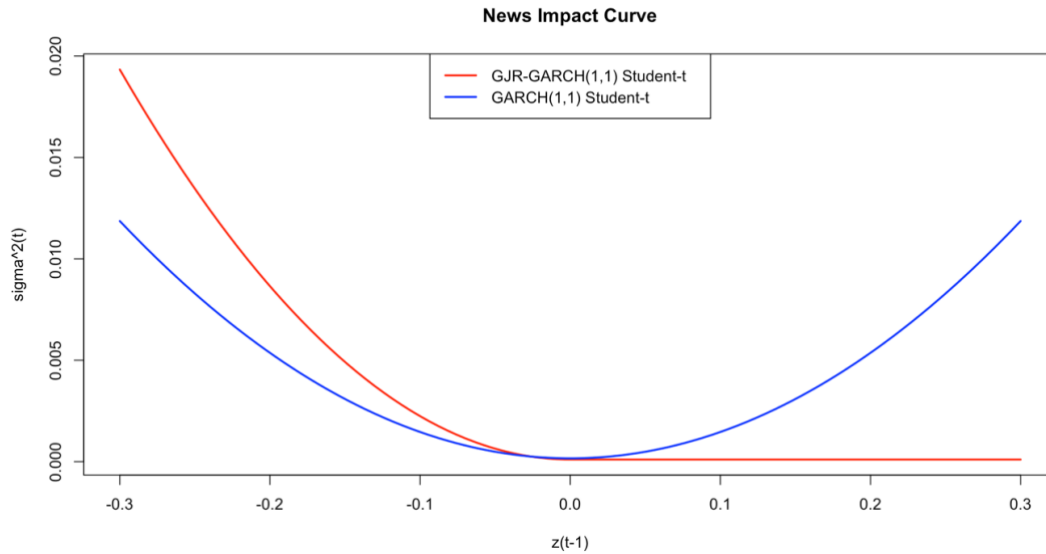


Figure 4: News Impact Curve - GARCH (1,1)-t vs GJR-GARCH (1,1)-t.

## 5.6 Economic Interpretation

The larger shock to volatility from negative returns has an economic explanation. This larger reaction is due to a "leverage" impact: when a firm's stock price falls (due to lower equity value), it becomes riskier. As a result, investors demand higher returns due to the increase in the firm's riskiness. Behavioral effects at the market level, such as panic sales, margin calls, or loss aversion could make these impacts even worse. The insignificance of the positive shock coefficient in the GJR-

GARCH equation indicates that short-term revisions in volatility are primarily caused by negative return shocks. The finite student-t degrees-of-freedom estimate of about 7.6 indicates that the tails are heavier tailed than a normal distribution and thus explains some of the extreme returns in March 2020.

## 6. Conclusion

This study uses symmetric and asymmetric GARCH-type models to capture the conditional volatility of AEX log returns from January 2010 through March 2023. We find strong evidence of the presence of a volatility-clustering effect and ARCH effects which support the rejection of constant variance. The GARCH (1,1) can capture a substantial degree of volatility persistence; however, our diagnostics indicate that there is a significant GJR-GARCH asymmetry parameter which supports the notion that volatility increases in response to a negative shock by a greater amount than it does due to positive shock of equal magnitude.

Our analysis demonstrates that the student-t GJR-GARCH (1,1) specification provides the best fit based on all examined information criteria. The likelihood ratio test rejects the symmetric restriction  $\gamma = 0$  in favor of the GJR-GARCH, supporting rejection of  $H_0$ . In addition, the estimation of a finite number of degrees of freedom for the student-t distribution supports the use of fat-tails over normal distributions.

In summary, this research shows that both fat tails and asymmetry play a role in capturing (AEX) return volatility. An extension to this research could include analyzing the forecasting ability of our estimated models in terms of their predictive power. We have identified two possible approaches including rolling window Value at Risk (VaR) back-testing of forecasts as well as developing and evaluating regime switching volatility models.

## Appendix

This appendix provides supplementary details on the estimation and diagnostics that support the analysis in the main text.

### A.1 ARMA Model Comparison

*Table A1: ARMA Specification Comparison*

| Specification | BIC    | Comment                                |
|---------------|--------|----------------------------------------|
| ARMA (0,0)    | -6.357 | constant mean                          |
| ARMA (1,1)    | -6.354 | coefficients near-cancel               |
| ARMA (2,2)    | -6.349 | overparameterized                      |
| ARMA (5,5)    | -6.330 | Severe redundancy: coefficients offset |

*Note: Lower BIC indicates better fit.*

### A.2 Volatility Persistence

Table A2 reports the persistence measure for each model.

*Table A2: Volatility Persistence*

| Model                     | $\alpha$ | $\beta$ | $\gamma$ | Persistence |
|---------------------------|----------|---------|----------|-------------|
| GARCH (1,1) Normal        | 0.0878   | 0.8991  | -        | 0.9869      |
| GARCH (1,1) Student-t     | 0.0836   | 0.9038  | -        | 0.9874      |
| GJR-GARCH (1,1) Student-t | 0.0225   | 0.9063  | 0.1097   | 0.9837      |

*Note: Persistence values close to 1 indicate that volatility shocks dissipate very slowly.*

### A.3 Diagnostic Summary of Preferred Model

The GJR-GARCH (1,1) specification of student-t errors passed all the standard tests for diagnostics at the conventional level of significance. Both standardized residuals and squared standardized residuals fail to reject the null of no residual autocorrelation at lags 1 through 25 (Ljung-Box Q-statistics,  $p > 0.05$ ). The ARCH-LM test also fails to reject the null of no remaining ARCH effects. The Jarque-Bera test still rejects normality, which is expected since the preferred model uses student-t innovations rather than normal innovations.

### A.4 Software and Reproducibility

In addition to providing the results of our analysis via tables and figures, we provide an R script which accompanies this paper. This R script includes all commands related to data loading, parameter estimation and testing diagnostics. All these commands can be combined to produce the same results as those presented in this document. Other R packages utilized include *tseries* (for performing ADF/KPSS/Jarque Bera tests); *forecast* (for creating auto.arima forecasts, acf/pacf plots);

moments (for calculating skewness/kurtosis); aTSA (for performing ARCH LM tests); and xts (for producing time series objects).

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